



Programme syllabus

[An accessible version of the syllabus can be found in the Course and programme directory.](#)

Master's Programme, Computer Science 120.0 credits

Masterprogram, datalogi

Valid for students admitted to the education from autumn 26 (HT - Autumn term; VT - Spring term).

This is a translation of the Swedish, legally binding, programme syllabus.

Programme objectives

The aim of the Master's programme is to provide a broad education in Computer Science with the possibility of deepening and specializing within the area of computer science, the method science for construction of computer programmes including theoretical foundations as well as the practical ability to develop products and systems which include computers and software.

The programme will provide the students with the requisites and abilities to participate and lead work within evaluation, development and implementation of new technology within the field of computer science.

Knowledge and understanding

For the master's degree from the programme Computer Science, the student shall

- show knowledge and understanding in the main area of the education, including both broad knowledge in the field as well as substantial in-depth knowledge in certain areas of the field, and in-depth insight into current research and development work
- show in-depth methodological knowledge in the main area of education.

Skills and abilities

For the master's degree from the programme Computer Science, the student shall

- show a good analytical problem solving ability
- show the ability to independently define and solve construction problems within computer science
- show the requisites and abilities to participate in and develop practices implemented in industry, maintenance and academic research
- show the ability to clearly present and discuss their conclusions and the knowledge and arguments on which they are based, both in national and international contexts, orally and in writing in dialogue with different groups.

Ability to make judgments and adopt a standpoint

For the master's degree from the programme Computer Science, the student shall

- be able to evaluate the quality of scientific studies and show a reflective and critical approach to scientific and non-scientific texts
- through self-development, retain his/her own professional ability during a professional career
- follow the discussion about technology in society and contribute to it.

Extent and content of the programme

The Master's programme in Computer Science is in the second cycle and comprises 120 ECTS credits, which, at normal study rate, corresponds to two years. The programme is given in English, but some elective courses are given in Swedish.

One of the following tracks must be taken

- Data Science
- Interaction Design
- Cognitive Systems
- Parallel computation
- Software Technology
- Theoretical Computer Science
- Visualization and Graphics

In some tracks there is a subtrack that must be selected. Under special circumstances an individual track can be approved by the programme director.

Eligibility and selection

Admission to the program requires fulfillment of basic eligibility, as well as the following requirements for special eligibility:

- Mathematics: four different subjects with a total of 28.5 credits. These subjects must include a course in Calculus in one variable, a course in Linear Algebra, a course in Probability Theory and Statistics, and a course in Discrete Mathematics.
- Computer Technology/Computer Science and Engineering/Computer Science/Information Technology: three different subjects with a total of 22.5 credits. These subjects must include a course in Object Oriented programming, a course in Algorithms and Data Structure and an in-depth course in Algorithmic Complexity.

Within the programme track, there may be courses with additional requirements for special qualifications. The requirements are stated under the heading "Special eligibility" in the syllabus. Examples include the following:

- Multivariable analysis is a special eligibility requirement for compulsory courses within the tracks Data Science and Cognitive systems.
- Human-computer interaction is a special entry requirement for compulsory courses within the Interaction Design track.
- Additional prerequisites may exist for some conditionally elective or recommended courses within the tracks.

The selection process is based on the following criteria: study results (grades from previous studies), motivation for the studies (awards, prizes and publications) and previous education. Prior education is evaluated based on the quality and suitability of the subjects that are relevant for the applied

programme. For international prior education, university ranking is also part of the merit rating. The merit rating is carried out on a scale of 1-75.

For students admitted to a 5-year engineering programme at KTH with a place guarantee for this programme, the above corresponds by the following courses within the respective programme. The course codes refer to the admission year 2022 on the 5-year engineering programme and may vary for other admission years.

Master of Science in Computer Engineering (CDATE)

Mandatory courses:

- SF1625 Calculus in one variable
- SF1624 Algebra and geometry
- SF1935 Probability theory and statistics with application to machine learning
- SF1688 Discrete Mathematics
- DD1337 Programming
- DD1338 Algorithms and data structures
- DD2350 Algorithms, data structures and complexity

Master of Science in Information Technology (CINTE)

Mandatory courses:

- SF1625 Calculus in one variable
- SF1624 Algebra and geometry
- SF1912 Probability theory and statistics
- SF1610 Discrete mathematics
- ID1018 Programming I
- ID1021 Algorithms and data structures

Conditionally elective course:

- DD2350 Algorithms, data structures and complexity **or** DD2352 Algorithms and Complexity

Master of Science in Media Technology (CMETE)

Mandatory courses:

- SF1625 Calculus in one variable
- SF1624 Algebra and geometry
- SF1919 Probability theory and statistics
- DD1318 Programming and scientific computing
- DD1320 Applied Computer Science

Conditionally elective courses:

- SF1662 Discrete Mathematics
- DD2352 Algorithms and complexity
- DD1380 Java programming for Python programmers

Master of Science in Technical Physics (CTFYS)

Mandatory courses:

- SF1673 Analysis in one variable
- SF1672 Linear algebra
- SF1922 Probability theory and statistics
- DD1331 Fundamentals of programming
- DD1327 Fundamentals of computer science

In addition, the following prior knowledge needs to be acquired as elective course:

- SF1679/SF1662/SF1610 Discrete mathematics
- DD2352 Algorithms and complexity **or** DD2350 Algorithms, data structures and complexity
- DD1380 Java programming for Python programmers

Master of Science in Technical Mathematics (CTMAT)

Mandatory courses:

- SF1673 Analysis in one variable
- SF1672 Linear algebra
- SF1918 Probability theory and statistics
- SF1679 Discrete Mathematics
- DD1331 Fundamentals of programming
- DD1380 Java programming for Python programmers
- DD1328 Fundamentals of Computer Science for Scientific Computing

In addition, the following prior knowledge needs to be acquired as an elective course:

- DD2352 Algorithms and complexity **or** DD2350 Algorithms, data structures and complexity.

Implementation of the education

Structure of the education

Each academic year consists of two semesters which are 20 weeks each, and each semester is further divided into two study periods.

During the first study year, mandatory courses are taken of a total of 30 ECTS credits, and one of these is the Programme Integrating Course, 2 ECTS credits, that is spread over two years.

In addition to the compulsory courses, it is required to follow a track within the master programme computer science. The compulsory courses in each track vary between 7,5-22.5 credits. In addition, each track has conditionally elective courses between 7,5-22,5 credits. The programme includes 28 credits of elective courses.

The programme is concluded by a degree project, 30 ECTS credits.

Courses

The programme is course-based. Lists of courses are included in appendix 1.

Grading system

Courses in the first and the second cycle are graded on a scale from A to F. A-E are passing grades, A is the highest grade. The grades pass (P) and fail (F) are used for courses under certain circumstances.

The grading scale is found in the course syllabus.

Conditions for participation in the programme

Participation requires admission to courses within the programme, and a course registration.

For further studies, special admission requirements for the course are to be fulfilled. Special admission requirements are listed in the respective course syllabus.

Degree project

The degree project is the final part of the education. The project work may begin when special admission requirements for the course are fulfilled.

Degree

This degree is entitled Master of Science (120 credits)/Teknologie masterexamen. The text on the degree certificate states the educational programme completed by the student.

Appendix 1: Course list

Appendix 2: Specialisations



Appendix 1: Course list

Master's Programme, Computer Science (TCSCM)

General Courses

Study year 1

Mandatory courses (32.0 credits)

Course code	Course name	Credits	Edu. level
DA2210	Introduction to the Philosophy of Science and Research Methodology for Computer Scientists	6.0 credits	Second cycle
DD2300	Program Integrating Course in Computer Science	2.0 credits	Second cycle
DD2380	Artificial Intelligence	6.0 credits	Second cycle
DD2395	Computer Security	6.0 credits	Second cycle
DD2440	Advanced Algorithms	6.0 credits	Second cycle
IK2218	Protocols and Principles of the Internet	6.0 credits	Second cycle

Supplementary information

Students who in a previous degree have read a course corresponding to DD2380, DD2395, IK2218 or DD2440 may apply to read a replacement course instead. The application is submitted to the master coordinator who, after reviewing the previously read course, helps the student to choose an appropriate replacement course. The course should be at least 6 credits, advanced level and broad in computer science. It must not be included in the student's track and the subject matter may not be close to any of the other compulsory courses.

The choice of replacement course must be approved by the programme director.

Students at KTH:

- from **CINTE** who have read any of the following courses can apply to read a replacement course: IK1203 Network and Communication, ID1214 Artificial Intelligence and Applications, IV1013 Introduction to Computer Security, IK2206 Internet Security and Privacy. Contact the master coordinator according to the instructions above.
- from **CDATE**, who started 2011 or earlier, have to take the mandatory course ID2200 Operating System, 6 credits, in the master's programme.

Study year 2

Mandatory courses (32.0 credits)

Course code	Course name	Credits	Edu. level
DD2300	Program Integrating Course in Computer Science	2.0 credits	Second cycle
DA231X	Degree Project in Computer Science and Engineering, Second Cycle	30.0 credits	Second cycle

Track, Data Science (CSDA)

Study year 1

Mandatory courses (7.5 credits)

Course code	Course name	Credits	Edu. level
DD2421	Machine Learning	7.5 credits	Second cycle

Conditionally elective courses

Course code	Course name	Credits	Edu. level
DD2417	Language Engineering	7.5 credits	Second cycle
DD2420	Probabilistic Graphical Models	7.5 credits	Second cycle
DD2424	Deep Learning in Data Science	7.5 credits	Second cycle
DD2477	Search Engines and Information Retrieval Systems	7.5 credits	Second cycle

Recommended courses

Course code	Course name	Credits	Edu. level
DH2322	Interactive Data Visualization	7.5 credits	Second cycle
DD2438	Artificial Intelligence and Multi Agent Systems	15.0 credits	Second cycle
DT2112	Speech Technology	7.5 credits	Second cycle
DT2119	Speech and Speaker Recognition	7.5 credits	Second cycle

Supplementary information

Prerequisites:

SF1626 Calculus in Several Variable, 7,5 credits (or equivalent).

Information:

DD2421 Machine Learning, can **not** be included in the track **if DD1420** Foundations of Machine Learning, has been taken before. In that case, the DD2421 must be replaced with a conditionally elective course within the track.

Information regarding conditionally elective courses

Mandatory courses for both subtracks:

- **DD2421** Machine Learning,
- **DD2430** Project Course in Data Science.

One of the subtracks must be chosen:

Subtrack 1: Machine Learning

Mandatory course:

- **DD2424** Deep Learning in Data Science

One of these courses must be taken:

- **DD2434** Machine Learning Advanced Course
- **DD2437** Artificial Neural Networks and Deep Architectures
- **DD2420** Probabilistic Graphical Models.

Subtrack 2: Natural Language Processing

Mandatory courses:

- **DD2477** Search Engines and Information Retrieval Systems
- **DD2417** Language Engineering.

Some courses may be given every two years.

Study year 2

Mandatory courses (7.5 credits)

Course code	Course name	Credits	Edu. level
DD2430	Project Course in Data Science	7.5 credits	Second cycle

Conditionally elective courses

Course code	Course name	Credits	Edu. level
DD2434	Machine Learning, Advanced Course	7.5 credits	Second cycle
DD2437	Artificial Neural Networks and Deep Architectures	7.5 credits	Second cycle

Recommended courses

Course code	Course name	Credits	Edu. level
DD2447	Statistical Methods in Applied Computer Science	6.0 credits	Second cycle
EL2320	Applied Estimation	7.5 credits	Second cycle
SF2940	Probability Theory	7.5 credits	Second cycle
DD2257	Visualization	7.5 credits	Second cycle
DD2368	Quantum Neural Networks	7.5 credits	Second cycle
DD2413	Social Robotics	7.5 credits	Second cycle
DD2423	Image Analysis and Computer Vision	7.5 credits	Second cycle
DD2610	Deep Learning, advanced course	7.5 credits	Second cycle
ID2222	Data Mining	7.5 credits	Second cycle
ID2223	Scalable Machine Learning and Deep Learning	7.5 credits	Second cycle
SF1811	Optimization	6.0 credits	First cycle
DD2361	Advanced Topics in Deep Learning in Biomedical Image Analysis	7.5 credits	Second cycle

Supplementary information

Prerequisites:

SF1626 Calculus in Several Variable, 7,5 credits (or equivalent).

Information regarding conditionally elective courses

Mandatory courses for both subtracks:

- **DD2421** Machine Learning,
- **DD2430** Project Course in Data Science.

One of the subtracks must be chosen.

Subtrack 1: Machine Learning

Mandatory course:

- **DD2424** Deep Learning in Data Science

One of these courses must be taken:

- **DD2434** Machine Learning Advanced Course
- **DD2437** Artificial Neural Networks and Deep Architectures
- **DD2420** Probabilistic Graphical Models.

Subtrack 2: Natural Language Processing

Mandatory courses:

- **DD2477** Search Engines and Information Retrieval Systems
- **DD2417** Language Engineering.

Some courses may be given every two years.

Track, Interaction Design (CSID)

Study year 1

Mandatory courses (7.5 credits)

Course code	Course name	Credits	Edu. level
DH2628	Interaction Design Methods	7.5 credits	Second cycle

Conditionally elective courses

Course code	Course name	Credits	Edu. level
DH2400	Physical Interaction Design and Realization	7.5 credits	Second cycle
DH2632	Human-Computer Interaction, Research Seminars	3.0 credits	Second cycle
DH2670	Haptics, Tactile and Tangible Interaction	7.5 credits	Second cycle
DM2731	AI for Learning	7.5 credits	Second cycle
DH2310	Extended Reality in Theory and Practice	7.5 credits	Second cycle
DM2588	Generative AI for Media Technology and Interaction Design	7.5 credits	Second cycle

Information regarding conditionally elective courses

At least **13,5 credits** (either 7,5+7,5 credits, or 7,5+6+3 credits, or 9+6 hp, or 6+6+3 hp) must be taken:

- **DH2400** Physical Interaction Design and Realization
- **DH2632** Human-Computer Interaction Research Seminars
- **DH2408** Evaluation Methods in Human-Computer Interaction
- **DM2588** Generative AI for Media Technology and Interaction Design
- **DH2670** Haptics, Tactile and Tangible Interaction
- **DT2140** Multimodal Interaction and Interfaces.
- **DM2731** AI for Learning.
- **DH2310** Extended Reality in Theory and Practice.
- **DM2730** Technology Enhanced Learning

Some courses may be given every two years.

Study year 2

Mandatory courses (9.0 credits)

Course code	Course name	Credits	Edu. level
DM2630	User Experience Design and Evaluation	9.0 credits	Second cycle

Conditionally elective courses

Course code	Course name	Credits	Edu. level
DH2408	Evaluation Methods in Human-Computer Interaction	6.0 credits	Second cycle
DM2730	Technology Enhanced Learning	7.5 credits	Second cycle
DT2140	Multimodal Interaction and Interfaces	7.5 credits	Second cycle

Information regarding conditionally elective courses

At least **13,5 credits** (either 7,5+7,5 credits, or 7,5+6+3 credits, or 9+6 hp, or 6+6+3 hp) must be taken:

- **DH2400** Physical Interaction Design and Realization
- **DH2632** Human-Computer Interaction Research Seminars
- **DM2588** Generative AI for Media Technology and Interaction Design
- **DH2670** Haptics, Tactile and Tangible Interaction
- **DH2408** Evaluation Methods in Human-Computer Interaction
- **DT2140** Multimodal Interaction and Interfaces.
- **DM2731** AI for Learning.
- **DH2310** Extended Reality in Theory and Practice.
- **DM2730** Technology Enhanced Learning.

Some courses may be given every two years.

Track, Cognitive Systems (CSCS)

Study year 1

Mandatory courses (7.5 credits)

Course code	Course name	Credits	Edu. level
DD2421	Machine Learning	7.5 credits	Second cycle

Conditionally elective courses

Course code	Course name	Credits	Edu. level
DD2410	Introduction to Robotics	7.5 credits	Second cycle
DD2423	Image Analysis and Computer Vision	7.5 credits	Second cycle
DT2151	Project in Conversational Systems	7.5 credits	Second cycle
DD2417	Language Engineering	7.5 credits	Second cycle
DT2112	Speech Technology	7.5 credits	Second cycle
DT2119	Speech and Speaker Recognition	7.5 credits	Second cycle
DD2424	Deep Learning in Data Science	7.5 credits	Second cycle
DD2419	Project Course in Robotics and Autonomous Systems	9.0 credits	Second cycle
DD2413	Social Robotics	7.5 credits	Second cycle

Supplementary information

Prerequisites:

SF1626 Calculus in Several Variable, 7,5 credits (or equivalent).

DD2421 Machine Learning, can **not** be included in the track if **DD1420** Foundations of Machine Learning, has been taken before. In that case, the DD2421 must be replaced with a conditionally elective course within the track.

DD2421 is mandatory for both subtracks.

Information regarding conditionally elective courses

One of the subtracks must be taken.

Subtrack 1: Vision and Robotics

Mandatory course:

- **DD2424** Deep Learning in Data Science

At least **one** must be taken:

- **DD2410** Introduction to Robotics
- **DD2419** Project Course in Robotics and Autonomous Systems.

At least **one** must be taken:

- **DD2423** Image Analysis and Computer Vision
- **DD2413** Social Robotics.

Subtrack 2: Conversational System

Mandatory courses:

- **DD2417** Language Engineering
- **DT2151** Project in Conversational Systems

At least **one** must be taken:

- **DT2112** Speech Technology
- **DT2119** Speech and Speaker Recognition.

Some courses may be given every two years.

Study year 2

Conditionally elective courses

Course code	Course name	Credits	Edu. level
DD2417	Language Engineering	7.5 credits	Second cycle
DT2151	Project in Conversational Systems	7.5 credits	Second cycle
DT2112	Speech Technology	7.5 credits	Second cycle
DT2119	Speech and Speaker Recognition	7.5 credits	Second cycle
DD2413	Social Robotics	7.5 credits	Second cycle
DD2423	Image Analysis and Computer Vision	7.5 credits	Second cycle
DD2419	Project Course in Robotics and Autonomous Systems	9.0 credits	Second cycle
DD2410	Introduction to Robotics	7.5 credits	Second cycle
DD2424	Deep Learning in Data Science	7.5 credits	Second cycle

Information regarding conditionally elective courses

Supplementary information

Prerequisites: SF1626 Calculus in Several Variables, 7,5 credits (or equivalent).

Information regarding conditionally elective courses

One of the subtracks must be taken.

Subtrack 1: Vision and Robotics

Mandatory course:

- **DD2424** Deep Learning in Data Science

At least one of these must be taken:

- **DD2419** Project Course in Robotics and Autonomous Systems
- **DD2410** Introduction to Robotics

At least one of these must be taken:

- **DD2423** Image Analysis and Computer Vision
- **DD2413** Social Robotics.

Subtrack 2: Conversational System

Mandatory courses:

- **DD2417** Language Engineering
- **DT2151** Project in Conversational Systems

At least **one** must be taken:

- **DT2112** Speech Technology
- **DT2119** Speech and Speaker Recognition.

Some courses may be given every two years.

Track, Software Technology (CSST)

Study year 1

Mandatory courses (7.5 credits)

Course code	Course name	Credits	Edu. level
DD2480	Software Engineering Fundamentals	7.5 credits	Second cycle

Conditionally elective courses

Course code	Course name	Credits	Edu. level
DD2528	Dependable Autonomous Systems	7.5 credits	Second cycle
DD2585	Programmable Society with Blockchains and Smart Contracts	7.5 credits	Second cycle
DD2459	Software Reliability	7.5 credits	Second cycle
DD2481	Principles of Programming Languages	7.5 credits	Second cycle
DD2525	Language-Based Security	7.5 credits	Second cycle
DD2557	Program Semantics and Analysis	7.5 credits	Second cycle
DD2460	Software Safety and Security	7.5 credits	Second cycle

Recommended courses

Course code	Course name	Credits	Edu. level
DD2421	Machine Learning	7.5 credits	Second cycle
DD2477	Search Engines and Information Retrieval Systems	7.5 credits	Second cycle
ID1217	Concurrent Programming	7.5 credits	First cycle

Information regarding conditionally elective courses

One of the subtracks must be chosen:

Subtrack 1: Programming Languages

Mandatory courses:

-**DD2481** Principles of Programming Languages.

-**ID2202** Compilers and Execution Environments

One course must be taken (DD2373 and DD2557 are given alternately every two years):

- **DD2373** Automata and Languages (given Spring 2028)
- **DD2557** Program Semantics and Analysis (given Spring 2027)
- **DD2525** Language-Based Security.

Subtrack 2: Software Engineering

One of these must be taken:

- **DD2489** Scalable software Development with Functional Programming
- **DD2529** Project Course on Operating Systems and Compiler Support for Security

At least two courses must be taken (DD2482 and DD2485 are given alternately every two years):

- **DD2443** Parallel and Distributed Computing
- **DD2459** Software Reliability
- **DD2460** Software Safety and Security
- **DD2481** Principles of Programming Languages
- **DD2525** Language-Based Security
- **DD2482** Automated Software Testing and DevOps
- **DD2585** Programmable Society (given every two years, given Spring 2027)
- **DD2528** Dependable Autonomous Systems (given every two years, given Autumn 2027).

Some courses may be given every two years.

Study year 2

Conditionally elective courses

Course code	Course name	Credits	Edu. level
DD2443	Parallel and Distributed Computing	7.5 credits	Second cycle
DD2482	Automated Software Testing and DevOps	7.5 credits	Second cycle
DD2489	Scalable software Development with Functional Programming	7.5 credits	Second cycle
ID2202	Compilers and Execution Environments	7.5 credits	Second cycle
DD2528	Dependable Autonomous Systems	7.5 credits	Second cycle
DD2585	Programmable Society with Blockchains and Smart Contracts	7.5 credits	Second cycle
DD2529	Project Course on Operating Systems and Compiler Support for Security	7.5 credits	Second cycle
DD2460	Software Safety and Security	7.5 credits	Second cycle

Recommended courses

Course code	Course name	Credits	Edu. level
ID1217	Concurrent Programming	7.5 credits	First cycle
DD2421	Machine Learning	7.5 credits	Second cycle
DD2458	Problem Solving and Programming under Pressure	9.0 credits	Second cycle
DD2477	Search Engines and Information Retrieval Systems	7.5 credits	Second cycle
IK2215	Advanced Internetworking	7.5 credits	Second cycle
IK2221	Networked Systems for Machine Learning	7.5 credits	Second cycle
IK2227	Network Systems with Edge or Cloud Datacenters	7.5 credits	Second cycle

Information regarding conditionally elective courses

One of the subtracks must be chosen:

Subtrack 1: Programming Languages

Mandatory courses:

-**DD2481** Principles of Programming Languages.

-**ID2202** Compilers and Execution Environments

One course must be taken (DD2373 and DD2557 are given alternately every two years):

- **DD2373** Automata and Languages (given Spring 2028)
- **DD2557** Program Semantics and Analysis (given Spring 2027)
- **DD2525** Language-Based Security.

Subtrack 2: Software Engineering

One of these must be taken:

- **DD2489** Scalable software Development with Functional Programming
- **DD2529** Project Course on Operating Systems and Compiler Support for Security

At least two courses must be taken (DD2482 and DD2485 are given alternately every two years):

- **DD2443** Parallel and Distributed Computing
- **DD2459** Software Reliability
- **DD2460** Software Safety and Security
- **DD2481** Principles of Programming Languages
- **DD2525** Language-Based Security
- **DD2482** Automated Software Testing and DevOps
- **DD2585** Programmable Society (given every two years, given Spring 2027)
- **DD2528** Dependable Autonomous Systems (given every two years, given Autumn 2027).

Some courses may be given every two years.

Track, Theoretical Computer Science (CSTC)

Study year 1

Conditionally elective courses

Course code	Course name	Credits	Edu. level
SF2940	Probability Theory	7.5 credits	Second cycle
SF2956	Topological Data Analysis	7.5 credits	Second cycle
DD2542	Seminars on Theoretical Computer Science, Algorithms and Complexity	7.5 credits	Second cycle
DD2452	Formal Methods	7.5 credits	Second cycle
SF2741	Enumerative Combinatorics	7.5 credits	Second cycle
DD2448	Foundations of Cryptography	7.5 credits	Second cycle
DD2459	Software Reliability	7.5 credits	Second cycle
SF2930	Regression Analysis	7.5 credits	Second cycle
SF2972	Game Theory	7.5 credits	Second cycle
DD2557	Program Semantics and Analysis	7.5 credits	Second cycle

Recommended courses

Course code	Course name	Credits	Edu. level
ID1217	Concurrent Programming	7.5 credits	First cycle

Information regarding conditionally elective courses

Mandatory course for all: **DD2467** Individual Project in Theoretical Computer Science 7,5 credits.

Information om villkorligt valfria kurser

All should take at least one of these courses:

- **SF2740** Graph Theory (will be given every two years, given in Autumn 2027),
- **SF2972** Game Theory
- **SF2940** Probability Theory,
- **SF2741** Enumerative Combinatorics (will be given every two years, given in Autumn 2027),
- **SF2930** Regression Analysis,
- **SF2956** Topological Data Analysis.

The student may propose a change to other courses in mathematics at advanced level and a total of at least 7,5 credits. The proposal is submitted to the Master Coordinator well in advance of the course start. The course change must be approved by the Programme Director. In addition, one of the sub-tracks must be selected:

Subtrack 1: Algorithms, Complexity and Cryptography

Mandatory course:

- **DD2448** Foundations of Cryptography.

At least one of must be taken:

- **DD2542** Seminars on Theoretical Computer Science, Algorithms and Complexity (will be given every two years, given in Autumn 2026),
- **DD2445** Complexity Theory (will be given every two years, given in Autumn 2027).

Subtrack 2: Formal Methods and Semantics

At least one must be taken:

- **DD2452** Formal Methods (will be given every two years, given in Autumn 2026),
- **DD2552** Seminars on Theoretical Computer Science, Programming Languages and Formal Methods (will be given every two years, given in Autumn 2027).

At least 7,5 credits must be taken:

- **DD2459** Software Reliability,
- **DD2557** Program Semantics and Analysis (will be given every two years, given in Spring 2027)
- **DD2460** Software Safety and Security
- **DD2373** Automata and Languages (will be given every two years, given in Spring 2026)
- **DD2443** Parallel and Distributed Computing.

Study year 2

Mandatory courses (7.5 credits)

Course code	Course name	Credits	Edu. level
DD2467	Individual Project in Theoretical Computer Science	7.5 credits	Second cycle

Conditionally elective courses

Course code	Course name	Credits	Edu. level
SF2972	Game Theory	7.5 credits	Second cycle
SF2956	Topological Data Analysis	7.5 credits	Second cycle
SF2940	Probability Theory	7.5 credits	Second cycle
SF2930	Regression Analysis	7.5 credits	Second cycle
SF2741	Enumerative Combinatorics	7.5 credits	Second cycle
DD2443	Parallel and Distributed Computing	7.5 credits	Second cycle
DD2542	Seminars on Theoretical Computer Science, Algorithms and Complexity	7.5 credits	Second cycle
DD2445	Complexity Theory	7.5 credits	Second cycle
DD2460	Software Safety and Security	7.5 credits	Second cycle
DD2552	Seminars on Theoretical Computer Science, Programming Languages and Formal Methods	7.5 credits	Second cycle

Recommended courses

Course code	Course name	Credits	Edu. level
DD2366	Open Quantum Systems	7.5 credits	Second cycle
DD2367	Quantum Computing for Computer Scientists	7.5 credits	Second cycle
DD2447	Statistical Methods in Applied Computer Science	6.0 credits	Second cycle
DD2458	Problem Solving and Programming under Pressure	9.0 credits	Second cycle

Supplementary information

Mandatory course for all:

DD2467 Individual Project in Theoretical Computer Science, 7,5 credits

Information regarding conditionally elective courses

Mandatory course for all: **DD2467** Individual Project in Theoretical Computer Science 7,5 credits.

Information om villkorligt valfria kurser

All should take at least one of these courses:

- **SF2740** Graph Theory (will be given every two years, given in Autumn 2027),
- **SF2972** Game Theory
- **SF2940** Probability Theory,
- **SF2741** Enumerative Combinatorics (will be given every two years, given in Autumn 2026),
- **SF2930** Regression Analysis,
- **SF2956** Topological Data Analysis.

The student may propose a change to other courses in mathematics at advanced level and a total of at least 7,5 credits. The proposal is submitted to the Master Coordinator well in advance of the course start. The course change must be approved by the Programme Director. In addition, one of the subtracks must be selected:

Subtrack 1: Algorithms, Complexity and Cryptography

Mandatory course:

- **DD2448** Foundations of Cryptography.

At least one of must be taken:

- **DD2542** Seminars on Theoretical Computer Science, Algorithms and Complexity (will be given every two years, given in Autumn 2026),
- **DD2445** Complexity Theory (will be given every two years, given in Autumn 2027).

Subtrack 2: Formal Methods and Semantics

At least one must be taken:

- **DD2452** Formal Methods (will be given every two years, given in Autumn 2026),
- **DD2552** Seminars on Theoretical Computer Science, Programming Languages and Formal Methods (will be given every two years, given in Autumn 2027).

At least 7,5 credits must be taken:

- **DD2459** Software Reliability,
- **DD2557** Program Semantics and Analysis (will be given every two years, given in Spring 2027)
- **DD2460** Software Safety and Security
- **DD2373** Automata and Languages (will be given every two years, given in Spring 2026)
- **DD2443** Parallel and Distributed Computing.

Track, Visualization and Interactive Graphics (CSVG)

Study year 1

Mandatory courses (7.5 credits)

Course code	Course name	Credits	Edu. level
DD2258	Introduction to Visualization, Computer Graphics and Image/Video Processing	7.5 credits	Second cycle

Conditionally elective courses

Course code	Course name	Credits	Edu. level
DH2310	Extended Reality in Theory and Practice	7.5 credits	Second cycle
DH2323	Computer Graphics and Interaction	6.0 credits	Second cycle
DH2650	Computer Game Design	6.0 credits	Second cycle
DH2322	Interactive Data Visualization	7.5 credits	Second cycle

Recommended courses

Course code	Course name	Credits	Edu. level
DD2356	Methods in High Performance Computing	7.5 credits	Second cycle
DD2424	Deep Learning in Data Science	7.5 credits	Second cycle

Information regarding conditionally elective courses

At least **16,5 credits** must be taken:

- **DD2257** Visualization
- **DH2322** Interactive Data Visualization)
- **DH2310** Extended Reality in Theory and Practice
- **DH2323** Computer Graphics and Interaction
- **DH2413** Advanced Graphics and Interaction
- **DH2650** Computer Game Design.

Study year 2

Mandatory courses (6.0 credits)

Course code	Course name	Credits	Edu. level
DD2470	Advanced Topics in Visualization and Computer Graphics	6.0 credits	Second cycle

Conditionally elective courses

Course code	Course name	Credits	Edu. level
DD2257	Visualization	7.5 credits	Second cycle
DH2413	Advanced Graphics and Interaction	9.0 credits	Second cycle

Recommended courses

Course code	Course name	Credits	Edu. level
DD2356	Methods in High Performance Computing	7.5 credits	Second cycle
DD2423	Image Analysis and Computer Vision	7.5 credits	Second cycle
DD2424	Deep Learning in Data Science	7.5 credits	Second cycle
DM2350	Human Perception for Information Technology	7.5 credits	Second cycle

Information regarding conditionally elective courses

At least **16,5 credits** must be taken:

- **DD2257** Visualization
- **DH2322** Interactive Data Visualization
- **DH2323** Computer Graphics and Interaction
- **DH2413** Advanced Graphics and Interaction
- **DH2650** Computer Game Design.
- **DH2310** Extended Reality in Theory and Practice.

Track, Parallel Computing (CSPC)

Study year 1

Conditionally elective courses

Course code	Course name	Credits	Edu. level
DD2356	Methods in High Performance Computing	7.5 credits	Second cycle
DD2370	Computational Methods for Electromagnetics	7.5 credits	Second cycle
IK2221	Networked Systems for Machine Learning	7.5 credits	Second cycle
DD2358	Introduction to High Performance Computing	7.5 credits	Second cycle
DD2363	Methods in Scientific Computing	7.5 credits	Second cycle
DD2365	Advanced Computation in Fluid Mechanics	7.5 credits	Second cycle
DD2421	Machine Learning	7.5 credits	Second cycle
DD2360	Applied GPU Programming	7.5 credits	Second cycle
DD2375	Project Course in High-Performance Computing	7.5 credits	Second cycle
DD2367	Quantum Computing for Computer Scientists	7.5 credits	Second cycle
DD2443	Parallel and Distributed Computing	7.5 credits	Second cycle
ID2223	Scalable Machine Learning and Deep Learning	7.5 credits	Second cycle
DD2368	Quantum Neural Networks	7.5 credits	Second cycle
DD2366	Open Quantum Systems	7.5 credits	Second cycle
DD2444	Project Course in Scientific Computing	7.5 credits	Second cycle
DD2257	Visualization	7.5 credits	Second cycle

Recommended courses

Course code	Course name	Credits	Edu. level
DD2437	Artificial Neural Networks and Deep Architectures	7.5 credits	Second cycle
IK2227	Network Systems with Edge or Cloud Datacenters	7.5 credits	Second cycle
DD2480	Software Engineering Fundamentals	7.5 credits	Second cycle
ID2203	Distributed Systems, Advanced Course	7.5 credits	Second cycle
CM2014	Simulation Methods in Medical Engineering	7.5 credits	Second cycle
DT2212	Music Acoustics	7.5 credits	Second cycle

Information regarding conditionally elective courses

One of the subtracks must be taken:

Subtrack 1: High Performance Computing

Mandatory courses within the subtrack:

- DD2358 Introduction to High Performance Computing
- DD2356 Methods in High Performance Computing
- DD2360 Applied GPU Programming.

One of the courses should be taken:

- DD2375 Project Course in High-Performance Computing
- DD2367 Quantum Computing for Computer Scientists
- DD2370 Computational Methods for Electromagnetics
- DD2443 Parallel and Distributed Computing
- IK2221 Networked Systems for Machine Learning
- ID2223 Scalable Machine Learning and Deep Learning.

Subtrack 2: Quantum Computing

Mandatory courses within the subtrack:

- DD2358 Introduction to High Performance Computing
- DD2367 Quantum Computing for Computer Scientists
- DD2368 Quantum Neural Networks.

One of the courses should be taken:

- DD2366 Open Quantum Systems
- DD2375 Project Course in High-Performance Computing
- DD2360 Applied GPU Programming
- DD2356 Methods in High Performance Computing.

Subtrack 3: Computational Science

Mandatory courses within the subtrack:

- DD2363 Methods in Scientific Computing
- DD2365 Advanced Computation in Fluid Mechanics
- DD2444 Project Course in Scientific Computing.

One of the courses should be taken:

- DD2421 Machine Learning
- DD2370 Computational Methods for Electromagnetics
- DD2360 Applied GPU Programming
- DD2257 Visualization.

Study year 2

Conditionally elective courses

Course code	Course name	Credits	Edu. level
DD2367	Quantum Computing for Computer Scientists	7.5 credits	Second cycle
DD2368	Quantum Neural Networks	7.5 credits	Second cycle
DD2375	Project Course in High-Performance Computing	7.5 credits	Second cycle
DD2360	Applied GPU Programming	7.5 credits	Second cycle
ID2223	Scalable Machine Learning and Deep Learning	7.5 credits	Second cycle
DD2443	Parallel and Distributed Computing	7.5 credits	Second cycle
DD2421	Machine Learning	7.5 credits	Second cycle
DD2370	Computational Methods for Electromagnetics	7.5 credits	Second cycle
DD2257	Visualization	7.5 credits	Second cycle
DD2444	Project Course in Scientific Computing	7.5 credits	Second cycle
DD2358	Introduction to High Performance Computing	7.5 credits	Second cycle
DD2356	Methods in High Performance Computing	7.5 credits	Second cycle
IK2221	Networked Systems for Machine Learning	7.5 credits	Second cycle
DD2363	Methods in Scientific Computing	7.5 credits	Second cycle
DD2365	Advanced Computation in Fluid Mechanics	7.5 credits	Second cycle

Recommended courses

Course code	Course name	Credits	Edu. level
DD2437	Artificial Neural Networks and Deep Architectures	7.5 credits	Second cycle
BB2280	Molecular Modeling	7.5 credits	Second cycle
DD2366	Open Quantum Systems	7.5 credits	Second cycle
DD2402	Advanced Individual Course in Computational Biology	6.0 credits	Second cycle
DD2435	Mathematical Modelling of Biological Systems	9.0 credits	Second cycle
EL2820	Modelling of Dynamical Systems	7.5 credits	Second cycle
SF2565	Program Construction in C++ for Scientific Computing	7.5 credits	Second cycle

Information regarding conditionally elective courses

One of the subtracks must be taken:

Subtrack 1: High Performance Computing

Mandatory courses within the subtrack:

- DD2358 Introduction to High Performance Computing
- DD2356 Methods in High Performance Computing
- DD2360 Applied GPU Programming.

One of the courses should be taken:

- DD2375 Project Course in High-Performance Computing
- DD2367 Quantum Computing for Computer Scientists
- DD2370 Computational Methods for Electromagnetics
- DD2443 Parallel and Distributed Computing
- IK2221 Networked Systems for Machine Learning
- ID2223 Scalable Machine Learning and Deep Learning.

Subtrack 2: Quantum Computing

Mandatory courses within the subtrack:

- DD2358 Introduction to High Performance Computing
- DD2367 Quantum Computing for Computer Scientists
- DD2368 Quantum Neural Networks.

One of the courses should be taken:

- DD2366 Open Quantum Systems
- DD2375 Project Course in High-Performance Computing
- DD2360 Applied GPU Programming
- DD2356 Methods in High Performance Computing.

Subtrack 3: Computational Science

Mandatory courses within the subtrack:

- DD2363 Methods in Scientific Computing
- DD2365 Advanced Computation in Fluid Mechanics
- DD2444 Project Course in Scientific Computing.

One of the courses should be taken:

- DD2421 Machine Learning
- DD2370 Computational Methods for Electromagnetics
- DD2360 Applied GPU Programming
- DD2257 Visualization.



Appendix 2: Specialisations

Master's Programme, Computer Science (TCSCM)

Track, Data Science (CSDA)

In our society, huge amounts of data are produced. This track Data Science is about methods for managing and analyzing data from various sources, e.g. images, video, text. Within the track one of two sub-tracks can be selected; Machine Learning or Natural Language Processing.

Track, Interaction Design (CSID)

The Interaction Design track deals with the development of interactive systems from a user-centered perspective in theory and practice. The track deals with concepts and models, methods for data collection and analysis, design and development and evaluation.

Track, Cognitive Systems (CSCS)

The track of Cognitive Systems is about the development of programs with artificial intelligence, i.e. abilities traditionally associated with people. Within the track you can choose to go deeper into robotics or conversational systems (systems that interact through human language, such as voice assistants, social robots and chat bots).

Track, Software Technology (CSST)

The track Software Technology is about methods for designing, developing and maintaining software. After a common course in Program Development Technology's basics, one of two sub-tracks is chosen: either Program Language specializing in Program Language basics, Compilers and Program Semantics or Software Engineering specializing in Large-scale Software Development, Automated Testing and Design of Reliable and Secure Software (distributed and/or autonomous).

Track, Theoretical Computer Science (CSTC)

The track Theoretical Computer Science is about the abstract and mathematical methods to study algorithms. Students learn to use formal methods and focus on verifiable properties of software and software systems. Within the track there are two sub-tracks, of which one must be selected, either Algorithms, Complexity and Cryptography or Formal Methods and Semantics.

Track, Visualization and Interactive Graphics (CSVG)

The Visualization and Graphics track stretches from basic visualization and graphics to modern research in the field. Visualization is mainly about making large and complex data understandable with the help of graphics, but has applications in computer games and other virtual environments.

Track, Parallel Computing (CSPC)

The Parallel Computing track focuses on advanced methods and techniques in High-Performance Computing (HPC) and Quantum Computing. Students learn to develop and optimize computational solutions on supercomputers, clusters equipped with cutting-edge GPUs and multi-core processors, as well as quantum computers and their simulators. These algorithms and methodologies are applied to accelerate computing across a wide range of fields, including deep learning, large language models, computational fluid dynamics, space physics, biological systems, and drug discovery. One of three subtracks, either High Performance Computing, Quantum Computing or Scientific Computing must be chosen.